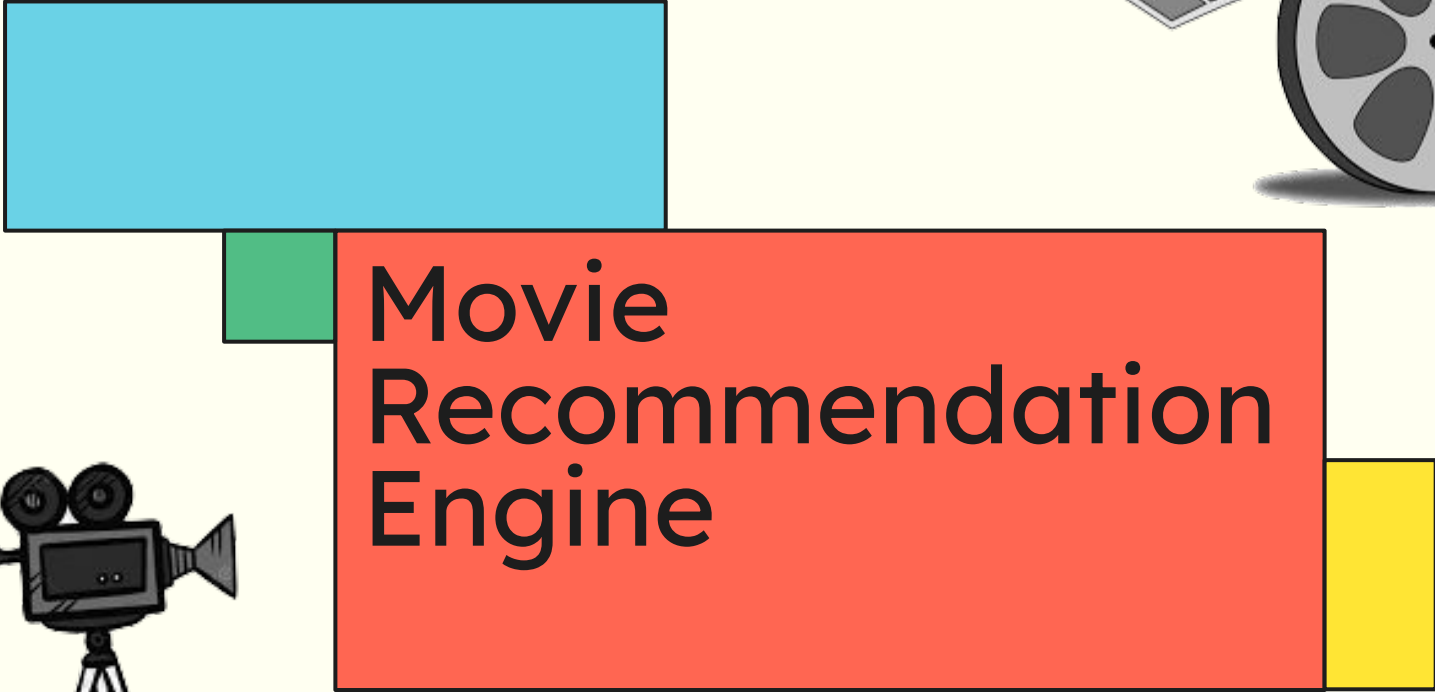
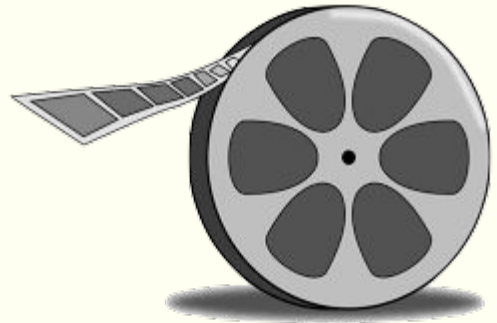




Movie Recommendation Engine

Pippa Hodgkins, Arielle Weinstein, Hartley Tran,
Ishan Monie, Hongyang Yu



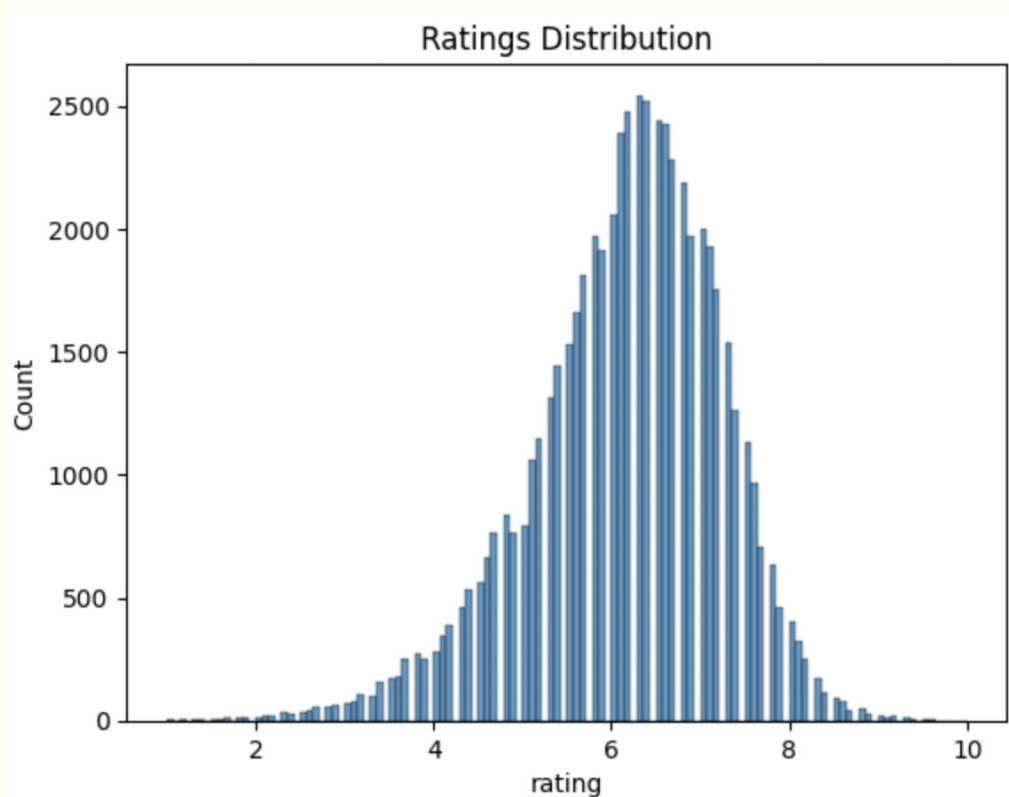
Overview and EDA

2

- Dataset contains ~60,000 movies
- Combined from MovieLens and IMDb datasets
- Features include ratings, votes, runtime, language, keywords, and revenue
- Used to build a movie recommendation system

Overview and EDA

3



Recommendation Engine

4

Features:

- Duration
- Rating
- Number of Ratings
- English vs. Non-English
- Keywords
- Worldwide gross
- Normalized rating
- Differential between rating and normalized rating



Engine:

- Movie based similarity (cosine similarity)
- Get top 50 most similar movies
- Sort by similarity and then the normalized rating to return most popular similar movies

Example Results

5



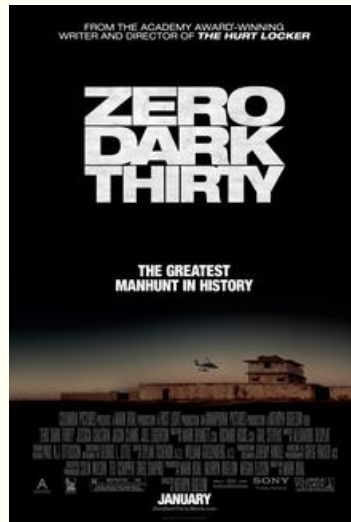
Monsters Inc.

Wish Dragon

Toy Story 3

Coco

Toy Story 2



The Post

I, Daniel Blake

Miss Sloane

Thirteen Days

Article 370

Clustering

6

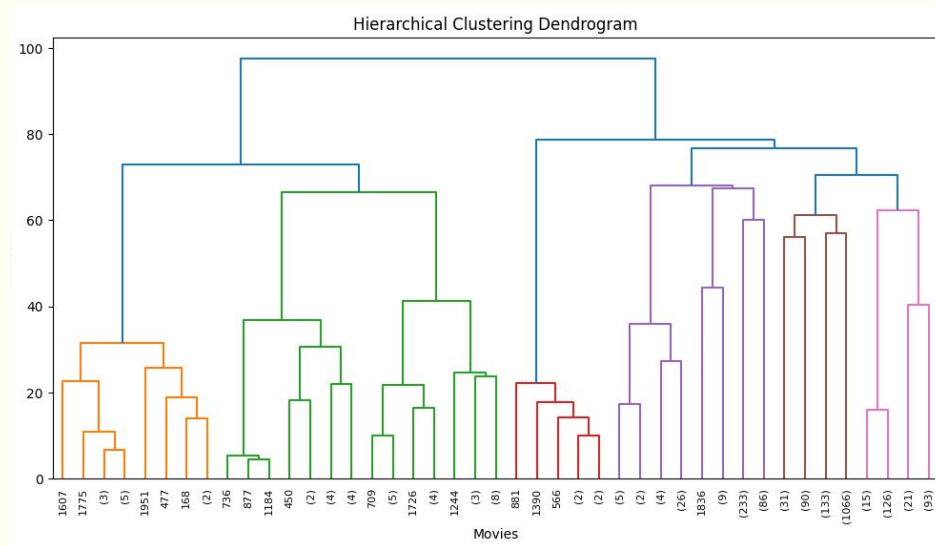
Features Used

- normalized_score (adjusted movie rating)
- log_votes (movie popularity)
- duration
- one-hot encoded genre indicators

Because the dataset contained many genre variables, PCA was applied to reduce dimensionality before clustering.

Method

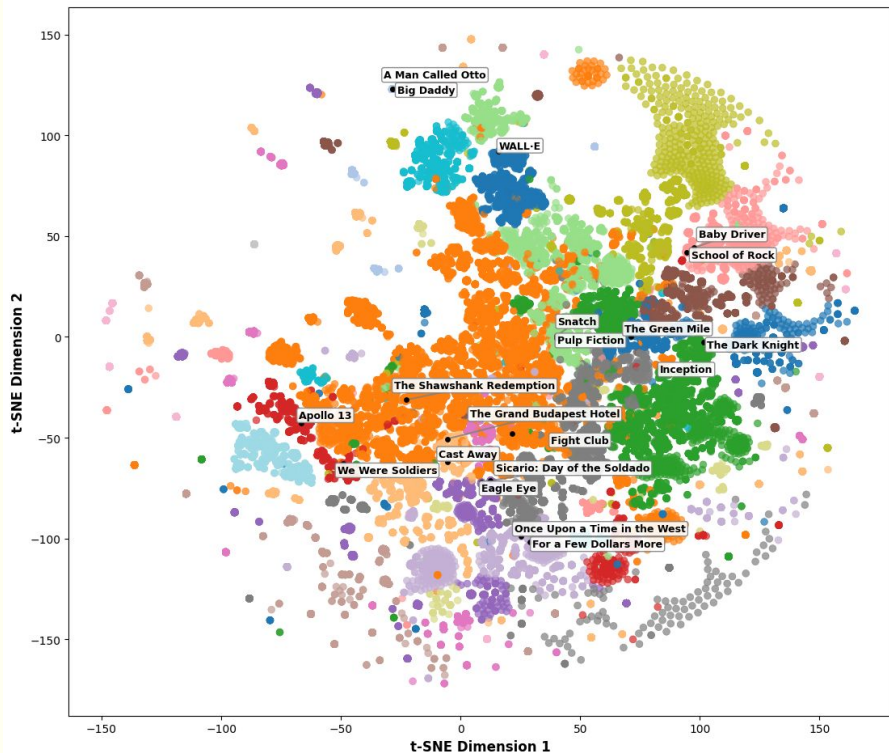
- Standardized all features
- Applied PCA (20 components) to reduce noise
- Used Agglomerative Hierarchical Clustering (Ward linkage)
- Examined cluster structure using a dendrogram



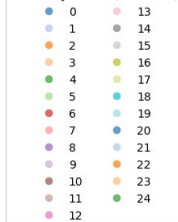
Clustering - t-SNE

7

Cinematic Taxonomies: 2D Map of Movie Neighborhoods



Taxonomy Branch (Cluster ID)



Cluster Breakdown

Cluster 0: Animated

Cluster 1: Western

Cluster 2: Action Blockbuster

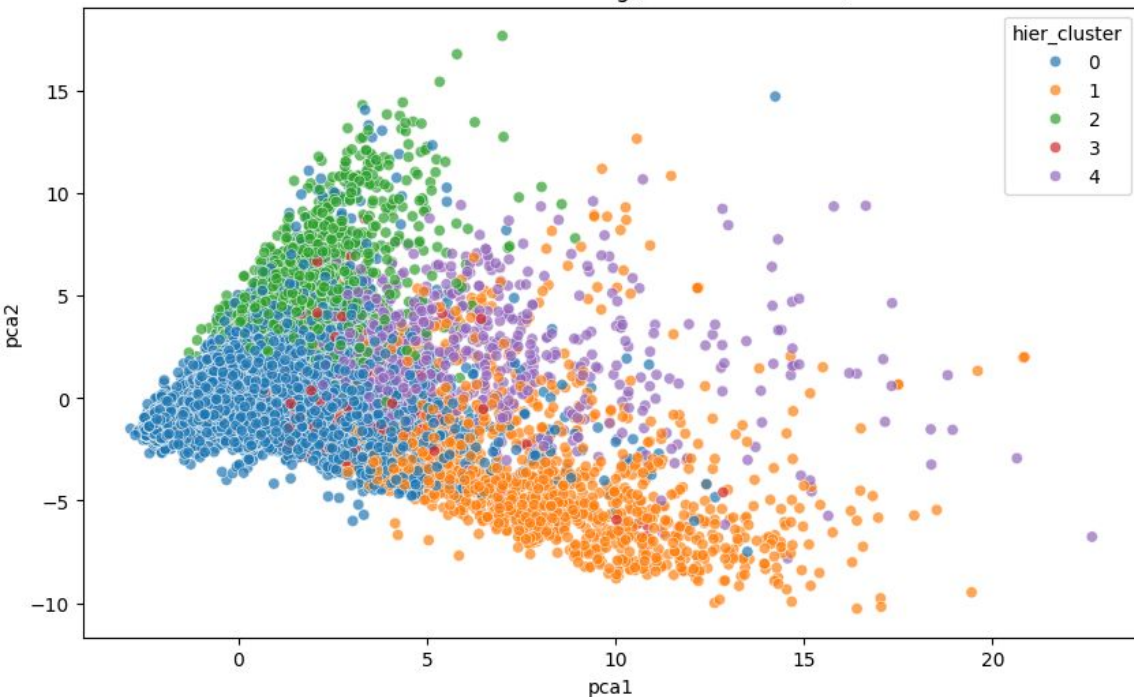
Cluster 3: Drama

Cluster 4: Sci-Fi

Clustering - PCA

8

Hierarchical Clustering (PCA Visualization)



Cluster Breakdown

Cluster 0: General Drama/Comedy (92%)

Cluster 1: Animated / Family Adventure (2.1%)

Cluster 2: Horror / Thriller (3.7%)

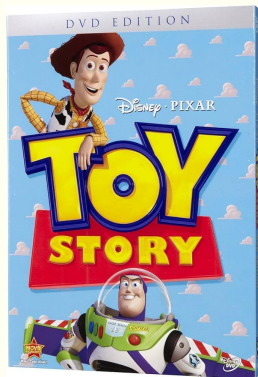
Cluster 3: Teen / Coming-of-Age Drama (0.9%)

Cluster 4: Sci-Fi / Action Blockbusters (1.3%)

While most films fall into a broad drama/comedy category, hierarchical clustering reveals some distinct genre based segments.

Example Results

9



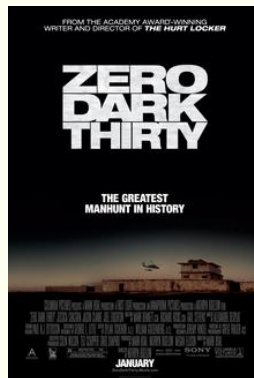
Toy Story 4

Toy Story 3

Cloudy with a Chance of Meatballs

Monsters, Inc.

Toy Story 2



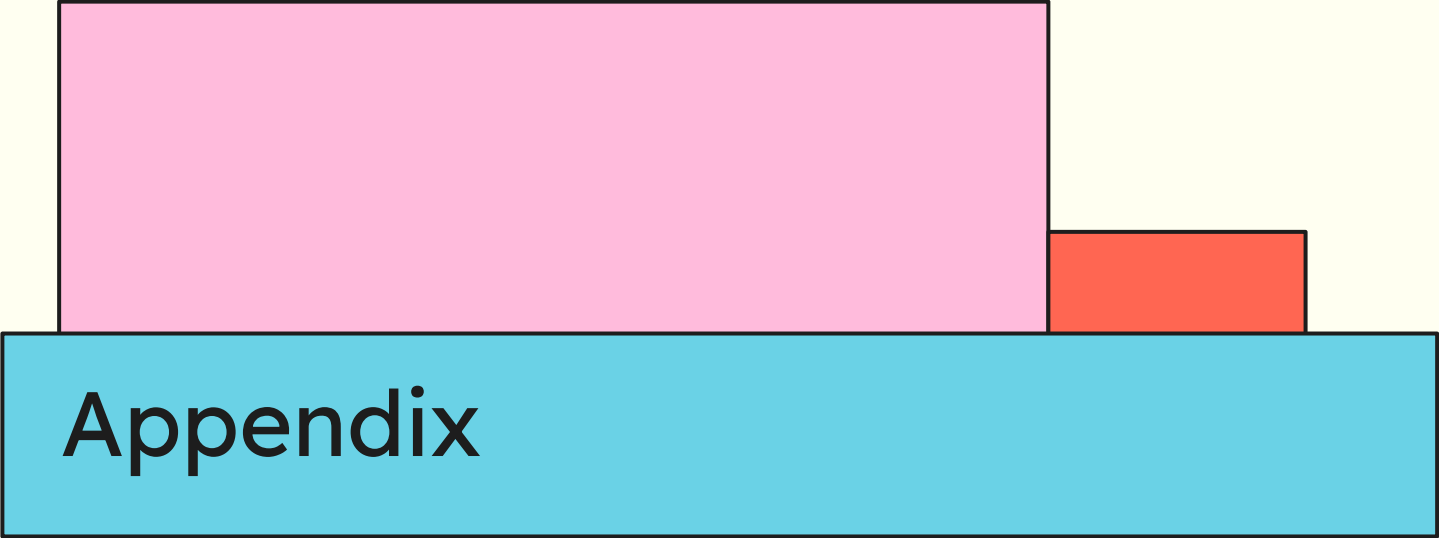
Thirteen Days

Bonhoeffer: Pastor,
Spy, Assassin

Argo

Bridge of Spies

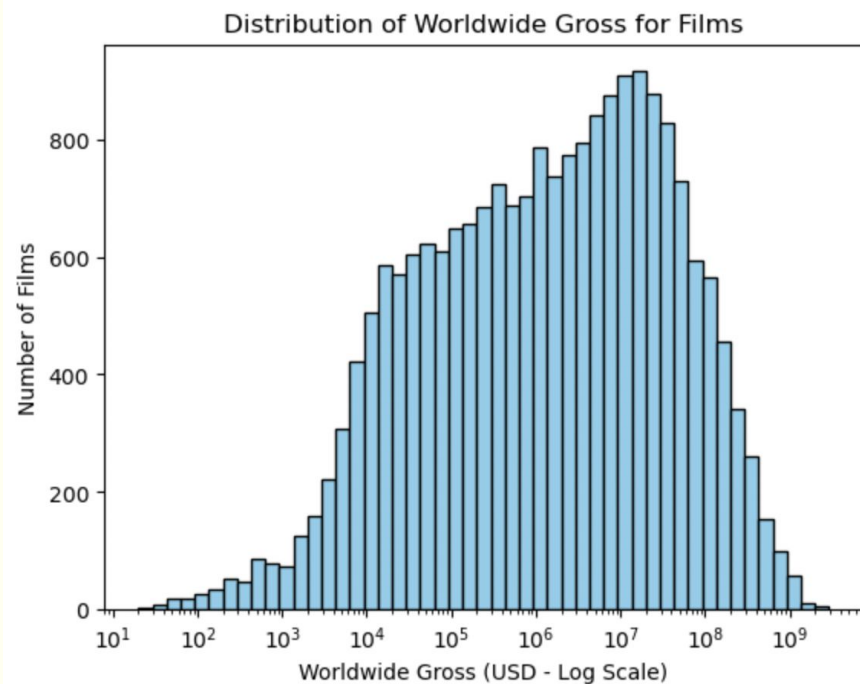
JFK



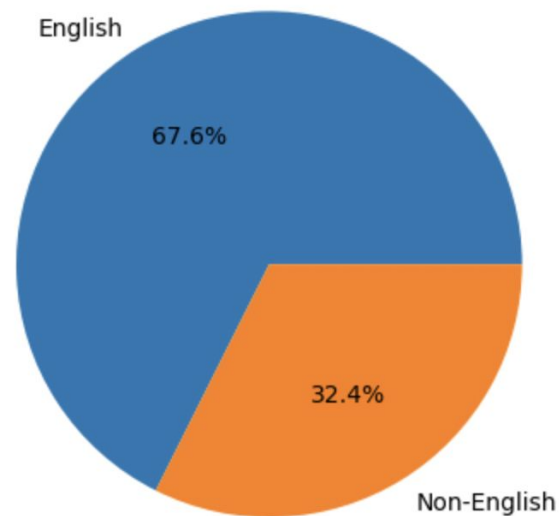
Appendix

Overview and EDA

11



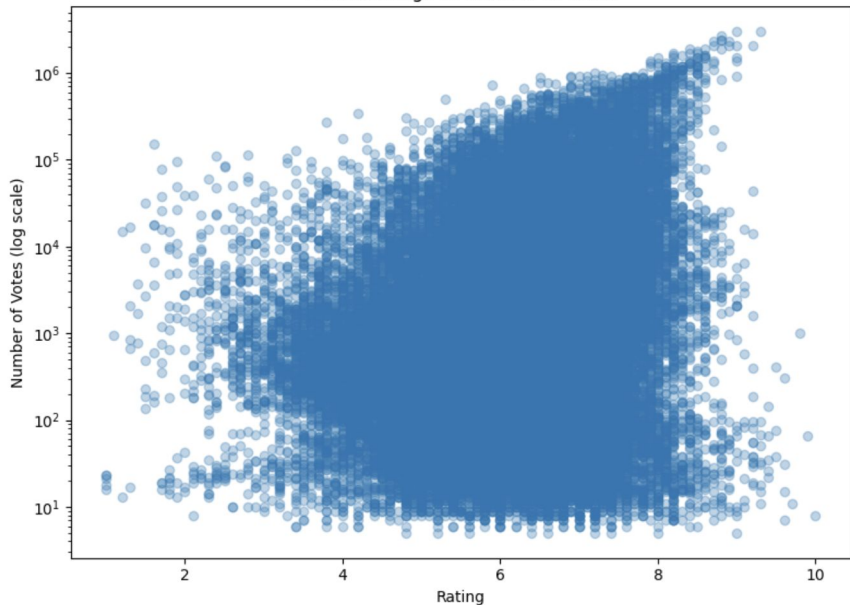
Distribution of English vs. Non-English Language Films



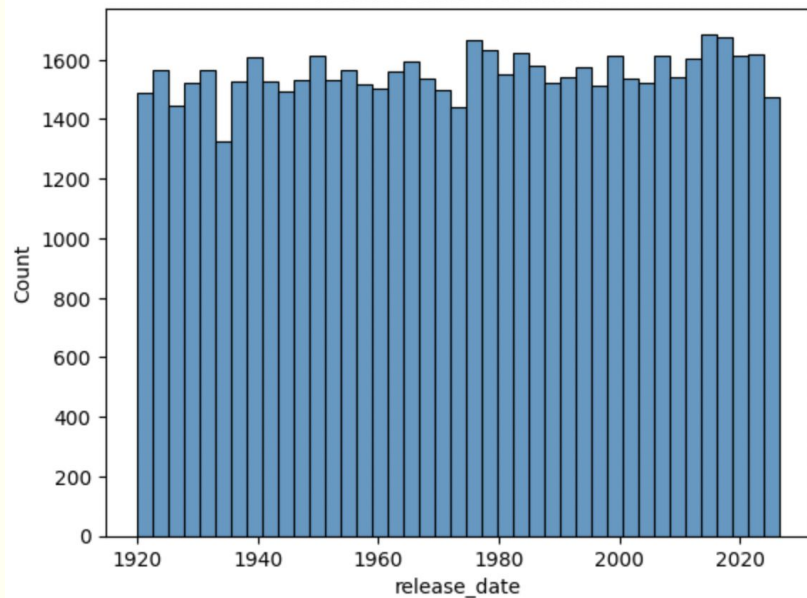
Overview and EDA

12

Movie Ratings vs Number of Votes



Release dates of movies



Data Cleaning

- Standardize data
- English and non english language vs. many languages
- Genres into dummy categories
- Filter keywords into most popular keywords
- Normalize number of ratings to account for movies that are rated less than others

Recommendation Engine cont'd

- Vectorization → in order to vectorize the keywords in order to incorporate them into the similarity score calculation, the tfidf module from sklearn was used to turn the keywords into vectors
 - These could then be utilized in the similarity calculation
- “Hybrid” structure of the recommender → the top 50 most similar movies are pulled, the similarity score is multiplied with the normalized rating and then divided by 10
 - This ensures that the recommender not only pulls similar movies, but the most popular of those similar movies